

Question 1.2.2 Choose two *different* words in the dataset with a magnitude (absolute value) of correlation higher than 0.2 and plot a scatter plot with a line of best fit for them. Please do not pick “outer” and “space” or “san” and “francisco”. The code to plot the scatter plot and line of best fit is given for you, you just need to calculate the correct values to `r`, `slope` and `intercept`.

Hint 1: It’s easier to think of words with a positive correlation, i.e. words that are often mentioned together. Try to think of common phrases or idioms.

Hint 2: Refer to [Section 15.2](#) of the textbook for the formulas. For additional past examples of regression, see Homework 9.

```
In [22]: word_x = 'kill'
        word_y = 'gun'

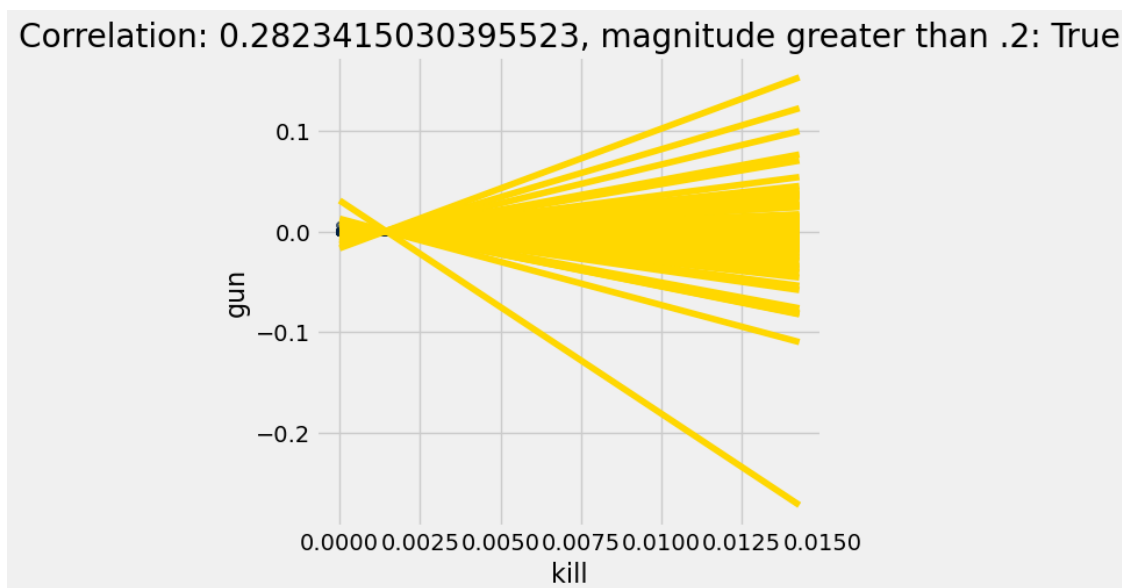
        # These arrays should make your code cleaner!
        arr_x = movies.column(word_x)
        arr_y = movies.column(word_y)

        x_su = (arr_x - np.mean(arr_x))/ np.std(arr_x)
        y_su = (arr_y - np.mean(arr_y))/ np.std(arr_y)

        r = np.mean(x_su * y_su)

        slope = r * (y_su/x_su)
        intercept = np.mean(arr_y) - slope * np.mean(arr_x)

        # DON'T CHANGE THESE LINES OF CODE
        movies.scatter(word_x, word_y)
        max_x = max(movies.column(word_x))
        plots.title(f"Correlation: {r}, magnitude greater than .2: {abs(r) >= 0.2}")
        plots.plot([0, max_x * 1.3], [intercept, intercept + slope * (max_x*1.3)], color='gold');
```

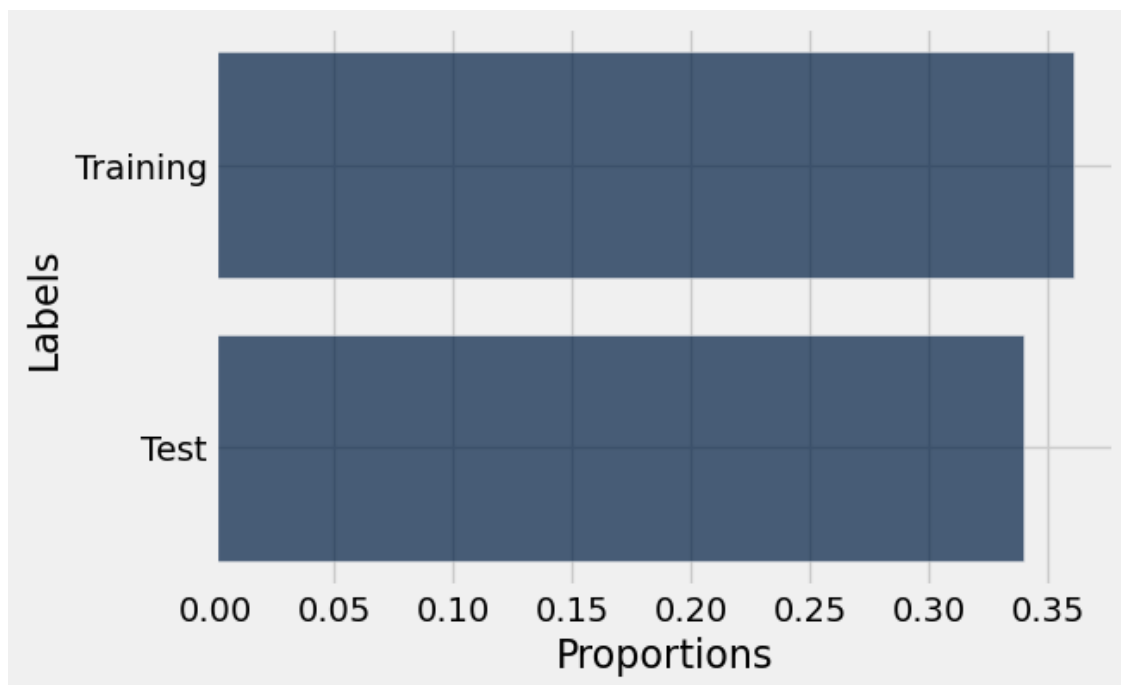


Question 1.3.1 Draw a horizontal bar chart with two bars that show the proportion of Comedy movies in each dataset (`train_movies` and `test_movies`). The two bars should be labeled “Training” and “Test”. Complete the function `comedy_proportion` first; it should help you create the bar chart.

Hint: Refer to [Section 7.1](#) of the textbook if you need a refresher on bar charts.

```
In [26]: def comedy_proportion(table):
    # Return the proportion of movies in a table that have the comedy genre.
    comedy = table.where("Genre", are.equal_to("comedy")).num_rows
    return comedy/table.num_rows

# The staff solution took multiple lines. Start by creating a table.
# If you get stuck, think about what sort of table you need for barh to work
labels = make_array('Training', 'Test')
proportions = make_array(comedy_proportion(train_movies), comedy_proportion(test_movies))
table_props = Table().with_columns('Labels', labels, 'Proportions', proportions)
table_props.barh(0,1)
```



Question 3.1.7 In two sentences or less, describe how you selected your features.

I selected the features by deciding on words that I thought would commonly appear for either comedy or thriller movies only. I opted for words that were not too close to the line because I want the word to more easily identify only one genre and not both.

Question 3.3.3

Do you see a pattern in the types of movies your classifier misclassifies? In two sentences or less, describe any patterns you see in the results or any other interesting findings from the table above. If you need some help, try looking up the movies that your classifier got wrong on Wikipedia.

Most of the movies that were classified as “correct” were thriller movies, and most of the movies that were classified as “not correct” were comedy movies.

Question 4.2

Do you see a pattern in the mistakes your new classifier makes? How good an accuracy were you able to get with your limited classifier? Did you notice an improvement from your first classifier to the second one? Describe in two sentences or less.

Hint: You may not be able to see a pattern.

The only pattern that I can see is that the majority of the few mistakes the classifier made were thriller movies. In terms of the improvement from my first classifier to the second one, I can see that the thriller movies were also classified more correctly in comparison to the first one, despite it having more incorrectly classified thriller movies than comedy movies.

Question 4.3

Given the constraint of five words, how did you select those five? Describe in two sentences or less.

I cut the ten words down to five by deciding on words that I thought would be more prominent and highly associated with either comedy or thriller movies only.

